

CODING CHALLENGE

A retail company receives daily order files from its sales application.

The source application uploads CSV files into an Azure Storage Account.

Snowflake should automatically load the new files whenever they arrive in Azure Storage.

The company wants:

1. A separate Snowflake warehouse for loading data
2. A separate user and role for data engineering
3. Data ingestion from Azure Storage Account
4. Initial load from CSV file
5. Automatic incremental loading when new CSV files are uploaded
6. Avoid duplicate records in the final table

Important Concept for Students

Snowflake does not automatically detect new rows added inside the same old file.

Best practice is:

Do not edit the already uploaded CSV file.

Instead, upload a new CSV file for every new batch.

Example:

retail_orders_initial.csv

retail_orders_incremental_20260508.csv

retail_orders_incremental_20260509.csv

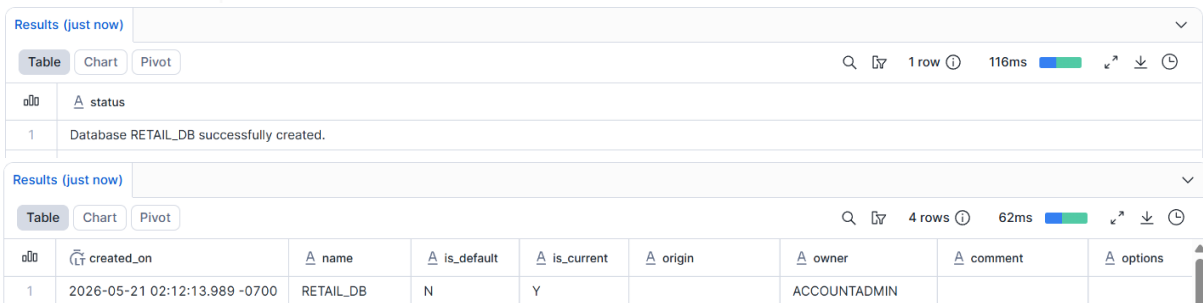
retail_orders_incremental_20260510.csv

Snowpipe detects and loads new files, not newly edited rows inside the same file.

Part A: Snowflake Setup

1. Create a database named RETAIL_DB.

1  CREATE DATABASE IF NOT EXISTS RETAIL_DB;



The screenshot displays two query result panels from the Snowflake interface. The top panel shows a single row with the message 'Database RETAIL_DB successfully created.' The bottom panel shows a table with details about the newly created database.

Results (just now)								
Table Chart Pivot								
1 row 116ms								
1	Database RETAIL_DB successfully created.							

Results (just now)								
Table Chart Pivot								
4 rows 62ms								
created_on	name	is_default	is_current	origin	owner	comment	options	
2026-05-21 02:12:13.989 -0700	RETAIL_DB	N	Y		ACCOUNTADMIN			

2. Create a schema named RETAIL_DB.ORDERS_DEMO.

3



```
CREATE SCHEMA IF NOT EXISTS RETAIL_DB.ORDERS_DEMO;
```

Results (just now)

Table Chart Pivot

1 row 40ms

id	status
1	Schema ORDERS_DEMO successfully created.

Results (just now)

Table Chart Pivot

3 rows 40ms

id	created_on	name	is_default	is_current	database_name	owner	comment
1	2026-05-21 02:28:33.698 -0700	INFORMATION_SCHEMA	N	N	RETAIL_DB		Views describing the conten
2	2026-05-21 02:13:05.409 -0700	ORDERS_DEMO	N	Y	RETAIL_DB	ACCOUNTADMIN	
3	2026-05-21 02:12:14.009 -0700	PUBLIC	N	N	RETAIL_DB	ACCOUNTADMIN	

3. Create a warehouse named RETAIL_LOAD_WH.

```
5 CREATE WAREHOUSE IF NOT EXISTS RETAIL_LOAD_WH
6 WAREHOUSE_SIZE = 'X-SMALL'
7 AUTO_SUSPEND = 60
8 AUTO_RESUME = TRUE
9 COMMENT = 'Dedicated warehouse for retail data loading';
```

Results (just now)

Table Chart Pivot

1 row 98ms

id	status
1	Warehouse RETAIL_LOAD_WH successfully created.

Results (just now)

Table Chart Pivot

1 row 76ms

id	name	state	type	size	# running	# queued	is_default	is_current
1	RETAIL_LOAD_WH	SUSPENDED	STANDARD	X-Small	0	0	N	Y

4. Create a role named RETAIL_DATA_ENGINEER_ROLE.

11



```
CREATE ROLE IF NOT EXISTS RETAIL_DATA_ENGINEER_ROLE;
```

Results (just now)

Table Chart Pivot

1 row 51ms

id	status
1	Role RETAIL_DATA_ENGINEER_ROLE successfully created.

Results (just now)

Table Chart Pivot

1 row 58ms

id	created_on	name	is_default	is_current	is_inherited	# assigned_to_users	# granted_to_roles	# granted_r
1	2026-05-21 02:14:58	RETAIL_DATA_ENGINEER	N	N	N	2	0	

5. Create a user named RETAIL_ENGINEER_USER.

```

13 CREATE USER IF NOT EXISTS RETAIL_ENGINEER_USER
14     PASSWORD = 'RetailEng@2026!'
15     DEFAULT_ROLE = RETAIL_DATA_ENGINEER_ROLE
16     DEFAULT_WAREHOUSE = RETAIL_LOAD_WH
17     DEFAULT_NAMESPACE = 'RETAIL_DB.ORDERS_DEMO'
18     MUST_CHANGE_PASSWORD = FALSE;

```

Results (just now)

Table Chart Pivot

1 row 305ms

id	status
1	User RETAIL_ENGINEER_USER successfully created.

Results (just now)

Table Chart Pivot

1 row 31ms

id	name	created_on	login_name	display_name	first_name	last_name	email	mins_to_un
1	RETAIL_ENGINEER_USER	2026-05-21 02:15:32	RETAIL_ENGINEER_USER	RETAIL_ENGINEER_USER	null	null	null	null

6. Grant required permissions to the role and user.

```

20 GRANT USAGE ON WAREHOUSE RETAIL_LOAD_WH TO ROLE RETAIL_DATA_ENGINEER_ROLE;
21 GRANT OPERATE ON WAREHOUSE RETAIL_LOAD_WH TO ROLE RETAIL_DATA_ENGINEER_ROLE;
22 GRANT USAGE ON DATABASE RETAIL_DB TO ROLE RETAIL_DATA_ENGINEER_ROLE;
23 GRANT USAGE ON SCHEMA RETAIL_DB.ORDERS_DEMO TO ROLE RETAIL_DATA_ENGINEER_ROLE;
24 GRANT CREATE TABLE ON SCHEMA RETAIL_DB.ORDERS_DEMO TO ROLE RETAIL_DATA_ENGINEER_ROLE;
25 GRANT CREATE STAGE ON SCHEMA RETAIL_DB.ORDERS_DEMO TO ROLE RETAIL_DATA_ENGINEER_ROLE;
26 GRANT CREATE PIPE ON SCHEMA RETAIL_DB.ORDERS_DEMO TO ROLE RETAIL_DATA_ENGINEER_ROLE;
27 GRANT CREATE STREAM ON SCHEMA RETAIL_DB.ORDERS_DEMO TO ROLE RETAIL_DATA_ENGINEER_ROLE;
28 GRANT CREATE TASK ON SCHEMA RETAIL_DB.ORDERS_DEMO TO ROLE RETAIL_DATA_ENGINEER_ROLE;
29 GRANT CREATE FILE FORMAT ON SCHEMA RETAIL_DB.ORDERS_DEMO TO ROLE RETAIL_DATA_ENGINEER_ROLE;
30 GRANT ALL ON ALL TABLES IN SCHEMA RETAIL_DB.ORDERS_DEMO TO ROLE RETAIL_DATA_ENGINEER_ROLE;
31 GRANT ALL ON FUTURE TABLES IN SCHEMA RETAIL_DB.ORDERS_DEMO TO ROLE RETAIL_DATA_ENGINEER_ROLE;
32
33 GRANT ROLE RETAIL_DATA_ENGINEER_ROLE TO USER RETAIL_ENGINEER_USER;
34 GRANT ROLE RETAIL_DATA_ENGINEER_ROLE TO USER MS7039;
35
36 GRANT EXECUTE TASK ON ACCOUNT TO ROLE RETAIL_DATA_ENGINEER_ROLE;

```

Results (just now)

Table Chart Pivot

1 row 35ms

id	status
1	Statement executed successfully.

Part B: Azure Storage Setup

7. Create an Azure Storage Account.

Home > snowflakeblobstorage1234_1779352724800 | Overview

snowflakeblobstorage1234 Storage account

Check data resiliency for storage accounts Harden this storage account Is this storage account vulnerable

Search Upload Open in Explorer Delete Move Refresh Open in mobile CLI / PS Feedback

Overview

- Activity log
- Tags
- Diagnose and solve problems
- Access Control (IAM)
- Data migration
- Events
- Storage browser
- Storage Mover
- Partner solutions
- Resource visualizer
- Data storage
- Security + networking
 - Networking
 - Front Door and CDN
 - Access keys
 - Shared access signature

Essentials

Resource group (move) : rg-azuser5791_mml-local-g7HV1

Location : centralindia

Subscription (move) : Learner Account

Subscription ID : 4116346b-5ded-4f3b-8387-f8d055802adc

Disk state : Available

Tags (edit) : Add tags

Performance : Standard

Replication : Locally redundant storage (LRS)

Account kind : StorageV2 (general purpose v2)

Provisioning state : Succeeded

Created : 5/21/2026, 2:10:01 PM

Properties Monitoring Capabilities (7) Recommendations (0) Tutorials Tools + SDKs

Blob service

Hierarchical namespace	Disabled
Default access tier	Hot
Blob anonymous access	Disabled
Blob soft delete	Enabled (7 days)
Container soft delete	Enabled (7 days)
Versioning	Disabled
Change feed	Disabled
NFS v3	Disabled

Security

Require secure transfer for REST API operations	Enabled
Storage account key access	Enabled
Minimum TLS version	Version 1.2
Infrastructure encryption	Disabled

Networking

Public network access	Enabled
Public network access scope	Enable from all networks

8. Create a container named retail-data.

New container ✕

Name *
retail-data ✓

Anonymous access level ⓘ
Private (no anonymous access) ▾

i The access level is set to private because anonymous access is disabled on this storage account.

Advanced ▾

Create Give feedback 🗨

Home > snowflakeblobstorage1234_1779352724800 | Overview > snowflakeblobstorage1234

snowflakeblobstorage1234 Containers

Search Add container Upload Refresh Delete Change access level Restore containers Edit columns

Search containers by prefix Only show active containers ▾

Showing all 2 items

☐	Name	Last modified	Anonymous access level	Lease state	Storage connector
☐	 \$logs	5/21/2026, 2:10:34 PM	Private	Available	- ...
☐	 retail-data	5/21/2026, 2:35:32 PM	Private	Available	- ...

Successfully created storage container ✕
Successfully created storage container 'retail-data'.

9. Inside the container, create this folder path:

raw/orders/

Upload blob

☐ Overwrite if files already exist

^ Advanced

Blob type ⓘ
Block blob

☒ Upload .vhd files as page blobs (recommended)

Blob size ⓘ
4 MiB

Access tier ⓘ
Hot (Inferred)

Upload to folder
raw/orders/

Blob index tags ⓘ

Key	Value

Encryption scope

☒ Use existing default container scope

☐ Choose an existing scope

10. Upload the initial CSV file:

retail_orders_initial.csv

retail-data

Container

+ Add Directory

↑ Upload

↻ Refresh

🗑 Delete

📄 Copy

📄 Paste

🔄 Rename

🔑 Acquire lease

🔑 Break lease

🔗 Edit columns

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

retail-data > raw > orders

Authentication method: Access key (Switch to Microsoft Entra user account)

Only show active blobs

Showing all 1 items

☐	Name	Last modified	Access tier	Blob type	Size	Lease state
☐	[-]					...
☐	retail_orders_initial.csv	5/21/2026, 2:38:52 PM	Hot (Inferred)	Block blob	1.47 KiB	Available ...

Part C: Snowflake External Stage

11. Create a file format for CSV.

```

6 CREATE OR REPLACE FILE FORMAT RETAIL_CSV_FORMAT
7     TYPE = 'CSV'
8     FIELD_DELIMITER = ','
9     RECORD_DELIMITER = '\n'
10    SKIP_HEADER = 1
11    FIELD_OPTIONALLY_ENCLOSED_BY = '"'
12    NULL_IF = ('NULL', 'null', '')
13    EMPTY_FIELD_AS_NULL = TRUE
14    TRIM_SPACE = TRUE
15    DATE_FORMAT = 'YYYY-MM-DD'
16    COMMENT = 'CSV format for retail order files';

```

Results (just now)

Table Chart Pivot 1 row 83ms

status
File format RETAIL_CSV_FORMAT successfully created.

Results (just now)

Table Chart Pivot 1 row 62ms

created_on	name	database_name	schema_name	type	owner	comment	format_options
2026-05-21 02:35:25	RETAIL_CSV_FORMAT	RETAIL_DB	ORDERS_DEMO	CSV	RETAIL_DATA_ENGINEER	CSV format for retail order files	["TYPE": "CSV"]

12. Create an Azure storage integration or external stage.

13. Create an external stage pointing to:

azure://<storage_account>.blob.core.windows.net/retail-data/raw/orders/

Home > snowflakeblobstorage1234

snowflakeblobstorage1234 | Shared access signature

Storage account

Search

Events

Storage browser

Storage Mover

Partner solutions

Resource visualizer

Data storage

Containers

Classic file shares

Queues

Tables

Security + networking

Networking

Front Door and CDN

Access keys

Shared access signature

Encryption

Microsoft Defender for Cloud

Give feedback

Allowed services

☒ Blob ☐ File ☐ Queue ☐ Table

Allowed resource types

☐ Service ☒ Container ☒ Object

Allowed permissions

☒ Read ☒ Write ☒ Delete ☒ List ☒ Add ☒ Create ☐ Update ☐ Process ☒ Immutability storage ☒ Permanent delete

Blob versioning permissions

☒ Enables deletion of versions

Allowed blob index permissions

☒ Read/Write ☒ Filter

Start and expiry date/time

Start 05/21/2026 2:54:54 PM

End 05/21/2026 6:09:54 PM

(UTC+05:30) Chennai, Kolkata, Mumbai, New Delhi

Allowed IP addresses

For example, 168.1.5.65 or 168.1.5.65-168.1.5.70

Allowed protocols

☒ HTTPS only ☐ HTTPS and HTTP

```

20 CREATE OR REPLACE STAGE AZURE_RETAIL_ORDERS_STAGE
21     URL = 'azure://snowflakeblobstorage1234.blob.core.windows.net/retail-data/raw/orders/'
22     CREDENTIALS = (
23         AZURE_SAS_TOKEN = 'sv=2026-02-06&ss=b&srt=co&sp=rwdlaciytfx&se=2026-05-21T12:39:54Z&st=2026-05-21T09:24:54Z&spr=https&sig=W3Pvd8yWTUEIbd5iX6NObL1qhFjnt2wnWxib7oMmjK%3D'
24     )
25     FILE_FORMAT = RETAIL_CSV_FORMAT
26     COMMENT = 'Azure Blob Storage stage for retail orders (SAS Token auth)';

```

Results (just now)								
TableChartPivot 1 row 150ms								
	status							
1	Stage area AZURE_RETAIL_ORDERS_STAGE successfully created.							

Results (just now)								
TableChartPivot 1 row 45ms								
	created_on	name	database_name	schema_name	url	has_credentials	has_encryption_key	owner
1	2026-05-21 02:44:30	AZURE_RETAIL_ORDER	RETAIL_DB	ORDERS_DEMO	azure://snowflakeblobs	Y	N	RETAIL_DA

14. List files from the stage.

```
30  LIST @AZURE_RETAIL_ORDERS_STAGE;
```

Results (just now)				
TableChartPivot 1 row 68ms				
	name	# size	md5	last_modified
1	azure://snowflakeblobstorage1234.blob.core.	1508	5804105d2cc314f10149ba1970e750ea	Thu, 21 May 2026 09:08:52 GMT

Part D: Initial Data Load

15. Create a raw landing table named ORDERS_RAW.

```
5  CREATE OR REPLACE TABLE ORDERS_RAW (
6      ORDER_ID          NUMBER,
7      CUSTOMER_ID       VARCHAR(20),
8      CUSTOMER_NAME     VARCHAR(100),
9      PRODUCT_ID        VARCHAR(20),
10     PRODUCT_NAME       VARCHAR(100),
11     CATEGORY           VARCHAR(50),
12     QUANTITY           NUMBER,
13     UNIT_PRICE         NUMBER(10,2),
14     TOTAL_AMOUNT       NUMBER(10,2),
15     ORDER_DATE         DATE,
16     STATUS             VARCHAR(30),
17     REGION             VARCHAR(50),
18     LOADED_AT          TIMESTAMP_NTZ DEFAULT CURRENT_TIMESTAMP(),
19     SOURCE_FILE        VARCHAR(500)
20 );
```

Results (just now)	
TableChartPivot 1 row 548ms	
	status
1	Table ORDERS_RAW successfully created.

16. Create a final table named ORDERS_FINAL.

```

22 CREATE OR REPLACE TABLE ORDERS_FINAL (
23     ORDER_ID          NUMBER          PRIMARY KEY,
24     CUSTOMER_ID       VARCHAR(20),
25     CUSTOMER_NAME     VARCHAR(100),
26     PRODUCT_ID        VARCHAR(20),
27     PRODUCT_NAME      VARCHAR(100),
28     CATEGORY          VARCHAR(50),
29     QUANTITY          NUMBER,
30     UNIT_PRICE        NUMBER(10,2),
31     TOTAL_AMOUNT      NUMBER(10,2),
32     ORDER_DATE        DATE,
33     STATUS            VARCHAR(30),
34     REGION            VARCHAR(50),
35     INSERTED_AT       TIMESTAMP_NTZ DEFAULT CURRENT_TIMESTAMP(),
36     UPDATED_AT        TIMESTAMP_NTZ DEFAULT CURRENT_TIMESTAMP()
37 );

```

Results (just now)

Table Chart Pivot 1 row 187ms

id	status
1	Table ORDERS_FINAL successfully created.

Results (just now)

Table Chart Pivot 2 rows 100ms

id	created_on	name	database_name	schema_name	kind	comment	cluster_by	# rows
1	2026-05-21 02:59:00.037 -0700	ORDERS_FINAL	RETAIL_DB	ORDERS_DEMO	TABLE			
2	2026-05-21 02:57:52.761 -0700	ORDERS_RAW	RETAIL_DB	ORDERS_DEMO	TABLE			

17. Load the initial file into ORDERS_RAW.

```

41 COPY INTO ORDERS_RAW (
42     ORDER_ID, CUSTOMER_ID, CUSTOMER_NAME, PRODUCT_ID, PRODUCT_NAME,
43     CATEGORY, QUANTITY, UNIT_PRICE, TOTAL_AMOUNT, ORDER_DATE, STATUS, REGION,
44     LOADED_AT, SOURCE_FILE
45 )
46 FROM (
47     SELECT
48         $1, $2, $3, $4, $5,
49         $6, $7, $8, $9, $10, $11, $12,
50         CURRENT_TIMESTAMP(),
51         METADATA$FILENAME
52     FROM @AZURE_RETAIL_ORDERS_STAGE/retail_orders_initial.csv
53 )
54 FILE_FORMAT = (FORMAT_NAME = RETAIL_CSV_FORMAT)
55 ON_ERROR = 'CONTINUE';

```

Results (just now)

Table Chart Pivot 1 row 3.3s

id	file	status	# rows_parsed	# rows_loaded	# error_limit	# errors_seen	first_error	first_error...
1	azure://snowflakeblobs	LOADED	15	15	15	0	null	

18. Insert data into ORDERS_FINAL.


```

57 MERGE INTO ORDERS_FINAL tgt
58 USING (
59     SELECT DISTINCT
60         ORDER_ID, CUSTOMER_ID, CUSTOMER_NAME, PRODUCT_ID, PRODUCT_NAME,
61         CATEGORY, QUANTITY, UNIT_PRICE, TOTAL_AMOUNT, ORDER_DATE, STATUS, REGION
62     FROM ORDERS_RAW
63 ) src
64 ON tgt.ORDER_ID = src.ORDER_ID
65 WHEN MATCHED THEN UPDATE SET
66     STATUS      = src.STATUS,
67     UPDATED_AT = CURRENT_TIMESTAMP()
68 WHEN NOT MATCHED THEN INSERT (
69     ORDER_ID, CUSTOMER_ID, CUSTOMER_NAME, PRODUCT_ID, PRODUCT_NAME,
70     CATEGORY, QUANTITY, UNIT_PRICE, TOTAL_AMOUNT, ORDER_DATE, STATUS, REGION
71 ) VALUES (
72     src.ORDER_ID, src.CUSTOMER_ID, src.CUSTOMER_NAME, src.PRODUCT_ID, src.PRODUCT_NAME,
73     src.CATEGORY, src.QUANTITY, src.UNIT_PRICE, src.TOTAL_AMOUNT,
74     src.ORDER_DATE, src.STATUS, src.REGION
75 );

```

Results (just now)		
Table	Chart	Pivot
1 row 1.9s		
#	# number of rows inserted	# number of rows updated
1	15	0

Part E: Incremental Loading

19. Create a Snowpipe object to load new files automatically into ORDERS_RAW.

```

5 CREATE OR REPLACE PIPE RETAIL_ORDERS_PIPE
6 AUTO_INGEST = FALSE
7 COMMENT = 'Snowpipe: loads CSV files from Azure into ORDERS_RAW (manual refresh)'
8 AS
9 COPY INTO ORDERS_RAW (
10     ORDER_ID, CUSTOMER_ID, CUSTOMER_NAME, PRODUCT_ID, PRODUCT_NAME,
11     CATEGORY, QUANTITY, UNIT_PRICE, TOTAL_AMOUNT, ORDER_DATE, STATUS, REGION,
12     LOADED_AT, SOURCE_FILE
13 )
14 FROM (
15     SELECT
16         $1, $2, $3, $4, $5,
17         $6, $7, $8, $9, $10, $11, $12,
18         CURRENT_TIMESTAMP(),
19         METADATA$FILENAME
20     FROM @AZURE_RETAIL_ORDERS_STAGE
21 )
22 FILE_FORMAT = (FORMAT_NAME = RETAIL_CSV_FORMAT);

```

Results (just now)		
Table	Chart	Pivot
1 row 212ms		
#	status	
1	Pipe RETAIL_ORDERS_PIPE successfully created.	

25  ALTER PIPE RETAIL_ORDERS_PIPE REFRESH;

Results (just now)	
Table	Chart Pivot
File	Status
 Query produced no results	

Results (just now)

TableChartPivot

1 row45ms

	created_on	name	database_name	schema_name	definition	owner	notification_channel	comment
1	2026-05-21 03:45:04	RETAIL_ORDERS_PIPE	RETAIL_DB	ORDERS_DEMO	COPY INTO ORDERS_R	RETAIL_DATA_ENGINE	null	Snowpipe:

Results (just now)	
Table	Chart Pivot
SYSTEM\$PIPE_STATUS('RETAIL_ORDERS_PIPE')	
1	{"executionState":"RUNNING","pendingFileCount":0,"pendingHistoryRefreshJobsCount":0}

Results (just now)

TableChartPivot

🔍📄1 row ⓘ561ms

📄	🔗 FILE_NAME	🔗 STATUS	🔗 ROW_COUNT	🔗 ERROR_COUNT	🕒 LAST_LOAD_TIME
1	retail_orders_initial.csv	Loaded	15	0	2026-05-21 04:54:41.673 -0700

20. Create a stream on ORDERS_RAW.

```

28 CREATE OR REPLACE STREAM ORDERS_RAW_STREAM
29 ON TABLE ORDERS_RAW
30 APPEND_ONLY = TRUE
31 COMMENT = 'Stream to capture new rows inserted into ORDERS_RAW by Snowpipe';

```

Results (just now)	
Table	Chart Pivot
status	
1	Stream ORDERS_RAW_STREAM successfully created.

21. Create a task to merge new rows from the stream into ORDERS_FINAL.

```

33 CREATE OR REPLACE TASK MERGE_ORDERS_TASK
34     WAREHOUSE = RETAIL_LOAD_WH
35     SCHEDULE = '5 MINUTE'
36     WHEN SYSTEM$STREAM_HAS_DATA('ORDERS_RAW_STREAM')
37 AS
38 MERGE INTO ORDERS_FINAL tgt
39 USING (
40     SELECT DISTINCT
41         ORDER_ID, CUSTOMER_ID, CUSTOMER_NAME, PRODUCT_ID, PRODUCT_NAME,
42         CATEGORY, QUANTITY, UNIT_PRICE, TOTAL_AMOUNT, ORDER_DATE, STATUS, REGION
43     FROM ORDERS_RAW_STREAM
44     WHERE METADATA$ACTION = 'INSERT'
45 ) src
46 ON tgt.ORDER_ID = src.ORDER_ID
47 WHEN MATCHED THEN UPDATE SET
48     STATUS = src.STATUS,
49     UPDATED_AT = CURRENT_TIMESTAMP()
50 WHEN NOT MATCHED THEN INSERT (
51     ORDER_ID, CUSTOMER_ID, CUSTOMER_NAME, PRODUCT_ID, PRODUCT_NAME,
52     CATEGORY, QUANTITY, UNIT_PRICE, TOTAL_AMOUNT, ORDER_DATE, STATUS, REGION
53 ) VALUES (
54     src.ORDER_ID, src.CUSTOMER_ID, src.CUSTOMER_NAME, src.PRODUCT_ID, src.PRODUCT_NAME,
55     src.CATEGORY, src.QUANTITY, src.UNIT_PRICE, src.TOTAL_AMOUNT,
56     src.ORDER_DATE, src.STATUS, src.REGION
57 );

```

Results (1 minute ago)

Table Chart Pivot

1 row 90ms

id	status
1	Task MERGE_ORDERS_TASK successfully created.

22. Upload another CSV file into Azure Storage.

retail-data ...

Container

Search

+ Add Directory Upload Refresh Delete Copy Paste Rename Acquire lease Break lease Edit columns

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

retail-data > raw > orders

Authentication method: Access key (Switch to Microsoft Entra user account)

Search blobs by prefix (case-sensitive)

Only show active blobs

Showing all 2 items

Name	Last modified	Access tier	Blob type	Size	Lease state
[-]					...
retail_orders_incremental_20260508.csv	5/21/2026, 4:58:21 PM	Hot (Inferred)	Block blob	594 B	Available
retail_orders_initial.csv	5/21/2026, 2:38:52 PM	Hot (Inferred)	Block blob	1.47 KiB	Available

Successfully deleted 1 blobs and directories.

Results (just now)

Table Chart Pivot

1 row 492ms

id	File	Status
1	retail_orders_incremental_20260508.csv	SENT

retail-data ...

Container

Search

+ Add Directory Upload Refresh Delete Copy Paste Rename Acquire lease Break lease Edit columns

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

retail-data > raw > orders

Authentication method: Access key (Switch to Microsoft Entra user account)

Search blobs by prefix (case-sensitive)

Only show active blobs

Showing all 4 items

Name	Last modified	Access tier	Blob type	Size	Lease state
[-]					...
retail_orders_incremental_20260508.csv	5/21/2026, 5:33:24 PM	Hot (Inferred)	Block blob	594 B	Available
retail_orders_incremental_20260509.csv	5/21/2026, 5:49:56 PM	Hot (Inferred)	Block blob	585 B	Available
retail_orders_incremental_20260510.csv	5/21/2026, 5:49:56 PM	Hot (Inferred)	Block blob	590 B	Available
retail_orders_initial.csv	5/21/2026, 2:38:52 PM	Hot (Inferred)	Block blob	1.47 KiB	Available

Results (just now)

TableChartPivot

🔍🔗3 rows273ms

	File	Status
1	retail_orders_incremental_20260508.csv	SENT
2	retail_orders_incremental_20260509.csv	SENT
3	retail_orders_incremental_20260510.csv	SENT

Results (just now)

TableChartPivot

🔍🔗1 row64ms

	status
1	Task MERGE_ORDERS_TASK is scheduled to run immediately.

23. Verify that new rows are automatically loaded into Snowflake.

```

1  SELECT FILE_NAME, STATUS, ROW_COUNT, ERROR_COUNT, LAST_LOAD_TIME
2  FROM TABLE(INFORMATION_SCHEMA.COPY_HISTORY(
3      TABLE_NAME=>'ORDERS_RAW',
4      START_TIME=>DATEADD(HOURS, -24, CURRENT_TIMESTAMP())
5  ))
6  ORDER BY LAST_LOAD_TIME DESC;
7
8  SELECT * FROM ORDERS_RAW;
9
10  SELECT * FROM ORDERS_FINAL
11  ORDER BY ORDER_ID;
12
13  ALTER PIPE RETAIL_ORDERS_PIPE REFRESH;
14
15  SELECT COUNT(*) FROM ORDERS_FINAL;
16

```

Initially;

Results (just now)

TableChartPivot

🔍🔗1 row549ms🌈🔗⬇️🕒

📄	# COUNT(*)
1	

15

After;

Results (just now)

Table

Chart

Pivot

🔍

📄

1 row ⓘ

50ms

↶ ↷

⬇

🕒

📄	# COUNT(*)
1	20

Results (2 minutes ago)

Table

Chart

Pivot

🔍

📄

1 row ⓘ

40ms

↶ ↷

⬇

🕒

📄	# COUNT(*)
1	30